

30 Days SQL Course Assignment

/* 1) Total Number of Patients

Write an SQL query to find the total number of patients across all hospitals.*/

```
SELECT SUM("Patients Count") AS Total_No_of_Patients  
FROM Hospital_Data;
```

/* 2) Average Number of Doctors per Hospital

Retrieve the average count of doctors available in each hospital.*/

```
SELECT AVG("Doctors Count") AS avg_Doctors_Per_Hospital  
FROM Hospital_Data;
```

/* 3) Top 3 Departments with the Highest Number of Patients

Find the top 3 hospital departments that have the highest number of patients.*/

```
SELECT "Department",SUM("Patients Count") AS total_patients  
FROM Hospital_data  
GROUP BY "Department"  
LIMIT 3;
```

/* 4) Hospital with the Maximum Medical Expenses

Identify the hospital that recorded the highest medical expenses.*/

```
SELECT "Hospital Name",MAX("Medical Expenses") AS max_medical_expenses
FROM Hospital_Data
GROUP BY "Hospital Name";
```

/* 5) Daily Average Medical Expenses

Calculate the average medical expenses per day for each hospital.*/

```
SELECT "Hospital Name",
AVG("Medical Expenses" / EXTRACT(DAY FROM "Discharge Date") - EXTRACT(DAY FROM
"Admission Date") + 1) AS daily_avg_medical_expenses
FROM Hospital_Data
GROUP BY "Hospital Name";
```

/* 6) Longest Hospital Stay

Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.*/

```
SELECT "Hospital Name",
MAX("Discharge Date"-"Admission Date") AS longest_stay
FROM Hospital_Data
gROUP BY "Hospital Name";
```

/* 7)Total Patients Treated Per City

Count the total number of patients treated in each city.*/

```
SELECT "Location",  
SUM("Patients Count") AS total_patients  
FROM Hospital_Data  
GROUP BY "Location";
```

/* 8) Average Length of Stay Per Department

Calculate the average number of days patients spend in each department*/

```
SELECT "Department",  
AVG("Discharge Date"-"Admission Date") AS avg_length_of_stay  
FROM Hospital_Data  
GROUP BY "Department";
```

/* 9) Identify the Department with the Lowest Number of Patients

Find the department with the least number of patients.*/

```
SELECT "Department",  
MIN("Patients Count") AS min_patients  
FROM Hospital_Data  
GROUP BY "Department";
```

/* 10) Monthly Medical Expenses Report

Group the data by month and calculate the total medical expenses for each month.*/

```
SELECT EXTRACT(MONTH FROM "Admission Date") AS months,  
SUM("Medical Expenses") AS total_medical_expenses  
FROM Hospital_data  
GROUP BY months  
ORDER BY months;
```